

Insulin Use Negatively Impacts Glioblastoma Patient Survival

Robert Edwards ^{1*}, Julia Noreck ^{2*}, Justin Bessette ², Jonathan Ardit ², Jacob Kaiserman ²,
Rainy Wortelboer², Mary Ann Nguyen ³, William Siwik ¹, Shiva Gautam ⁴, Tara Szymanski ¹, Eric
T Wong ^{1,2}

¹ Brown University Health Cancer Institute, Rhode Island Hospital, Providence, Rhode Island

² The Warren Alpert Medical School of Brown University, Providence, Rhode Island

³ Department of Data Science, Boston University, Boston, Massachusetts

⁴ Department of Biostatistics, University of Florida at Gainesville, Gainesville, Florida

*These authors contributed equally to this work

Short title: Insulin Effects on Glioblastoma

Keywords: Insulin, Glioblastoma, Survival, Prognostic Factors

Address for correspondence:

Eric T Wong, MD, MA

Division of Hematology/Oncology, George 315

Brown University Health Cancer Institute & Rhode Island Hospital

593 Eddy Street

Providence, RI 02903

Tel: 401-444-4564

Fax: 401-444-4184

Email: ewong1@brownhealth.org or eric_wong@brown.edu

ORCID: 0000-0001-7112-6207

Additional Information

Acknowledgements: We thank Melanie Ho and Elizabeth Xu for help in additional data extraction from the Brown University Health Cancer Registry.

Authors' contributions: RE, JN, JB, JA, and ETW helped with data collection, data analysis, and writing the manuscript. JK, RW, WS and TS helped with data collection. MAN and SG helped with data analysis.

Ethics approval and consent to participate: The analysis of the patient database at Brown University Health Cancer Institute was performed in accordance with the Declaration of Helsinki and under an institutional review board approved protocol #218622.

Consent for publication: The data published in this manuscript contain no identifiable patient information or individual details, ensuring that patient anonymity is preserved.

Data availability: All data were analyzed using R statistical software. A code repository for all scripts used to generate figures in this publication can be found at this link:

<https://github.com/roberthedwardsIV/GBM-Diabetes-Analysis-Figures-Tables>

Competing Interest: The authors declare no conflict of interest.

Funding information: This research was funded in part by the *A Reason To Ride* research fund. JN received a 2023 Summer Research Assistantship from The Warren Alpert Medical School. JB and JA each received a T35 Short-Term Research Training Grant from the National Institutes of Health.

Abstract

Background: The prevalence of type 2 diabetes mellitus (T2DM) is increasing in the general population but its influence on glioblastoma patient survival is unclear.

Methods: We retrospectively analyzed the Cancer Center Registry at Brown University Health and found glioblastoma patients with (n=142) and without (n=715) T2DM who presented from 2005-2022.

Results: Patients with T2DM were older, with lower neurological performance and a lower rate of resection compared to those without. Median overall survival (OS) for diabetics was 7.9 compared to 11.9 months in non-diabetics ($p < 0.0001$). Karnofsky Performance Status and extent of tumor resection were significant prognostic factors. Diabetics who used insulin had a significantly shorter median OS than non-diabetics ($p < 0.0001$) or diabetics who did not use insulin ($p = 0.0109$). Dose matters for patients who used insulin, with a median OS of 5.2 versus 10.5 months for those who used a daily dose of >10 and <10 units, respectively ($p = 0.0006$). There was no correlation between tumor size and daily (Pearson $r^2 = 0.0025$, $p = 0.6846$) or cumulative (Pearson $r^2 = 0.0011$, $p = 0.7859$) insulin use.

Conclusions: Glioblastoma patients with T2DM using insulin had significantly compromised survival compared to non-diabetics or non-insulin-using diabetics. Attention to diabetes and insulin use will be important in the overall management of the glioblastoma population.

Introduction

Glioblastoma is the most prevalent and aggressive primary malignant brain tumor in adults ¹. Despite significant research endeavors over past decades ², prognosis remains poor with a 5-year survival rate below 10% ³. Glioblastoma survival may be influenced by multiple medical co-morbidities including type 2 diabetes mellitus (T2DM) ^{4,5}. It has been shown that the prevalence of T2DM has been rapidly increasing and it has become a major worldwide public health problem ⁶. Published research indicates that it is also linked to a higher risk of several malignancies including renal, pancreatic, cervical and uterine cancers ⁷. However, there is conflicting evidence for an association with glioblastoma. One study revealed that T2DM was causally linked with an increased risk ⁴, while others have found no association ^{8,9}. Remarkably, one meta-analysis suggested a decreased risk of gliomas in patients with T2DM ¹⁰.

As the prevalence of T2DM rises, it is critical to determine the survival impact of this metabolic disorder in glioblastoma patients. T2DM is related to aging, and the prevalence increases from 3.0% among those at age 18 to 44, to 14.5% in those 45 to 64, and 24.4% in those 65 or older ¹¹. Furthermore, the global prevalence of prediabetes, defined as impaired glucose tolerance or abnormal fasting glucose level, was 9.1% in 2021, and it is projected to increase to 10.0% by 2045 ¹². Therefore, T2DM and prediabetes are prevalent medical co-morbidities in the glioblastoma population. Indeed, about 11 to 26% of these patients have concurrent T2DM, and the use of dexamethasone to treat cerebral edema may trigger or further exacerbate existing hyperglycemia ^{13–15}. Here, we performed a retrospective analysis of the effect of T2DM and insulin on the glioblastoma population at Brown University Health.

Methods

Data collection

Between January 2005 and December 2022, 867 glioblastoma patients received treatment at Brown University Health Cancer Institute or Rhode Island Hospital, including those initially diagnosed at a different facility or eventually transferred elsewhere. General analysis of medical comorbidities on their survival has been published ⁵. Here, we specifically focused on the effect of T2DM on the entire cohort. Of the 867 patients identified, 142 had concurrent diabetes based on their ICD-9 or -10 codes from notes on or before the date of glioblastoma diagnosis (Table 1). All identified patients with glioblastoma and diabetes were included in this analysis.

Diagnosis date, death or censoring date, patient demographics, and clinical data, including age, Karnofsky Performance Status (KPS), *O*⁶-*Methylguanine-DNA methyltransferase* (*MGMT*) promoter methylation, isocitrate dehydrogenase (IDH) mutation status, dates and types of treatment, and extent of tumor resection were tabulated from the electronic health record. When not available in physician notes, the KPS at the time of diagnosis was estimated based on the clinical description of the patient. Medical co-morbidities and diabetic complications were documented according to ICD-9 or -10 codes from notes on or before the date of diagnosis.

Overall survival (OS) was calculated from diagnosis to death, rounded to the nearest whole number. Patients who could not be confirmed as deceased were censored at their last date of contact.

Statistical Analysis

Statistical analyses were performed in R version 4.4.1. For each co-morbidity and treatment analyzed, survival outcomes over time were determined using a Kaplan-Meier estimator (R packages: *survival*, *survminer*). Kruskal-Wallis and pairwise Wilcoxon Rank Sum tests were performed to assess differences among groups (R functions: *kruskal.test*, *pairwise.wilcox.test*), and chi-squared statistics (R function: *chisq.test*) were performed on

categorical variables between groups. Multivariate Cox proportional hazard models were performed to assess differences in survival probabilities (R package: *coxph*). Statistical significance was defined at $p \leq 0.05$.

To identify the insulin dose cutoff associated with differences in survival among patients taking insulin, the patient population was dichotomized using an iterative processing algorithm based on ascending insulin doses. The Python *lifelines* survival analysis library¹⁶ was used to generate a Kaplan-Meier estimator to compare the survival difference between low and high insulin groups. The low insulin group was defined as patients using an insulin dose < the cutoff, and the high insulin group was defined as patients utilizing an insulin dose $\geq$ the cutoff, at each insulin dose. The p value associated with this survival difference was then estimated.

Glioblastoma growth velocity measurements were performed according to the Response Assessment in Neuro-Oncology (RANO) criteria¹⁷. This is based on a 2-dimensional measurement of tumor size by determining the product of the 2 greatest perpendicular diameters of the tumor, using the post-gadolinium-enhanced T1-weighted Magnetization-Prepared Rapid Gradient Echo (MP RAGE) sequence. To further account for the broader infiltration of tumor cells beyond areas of enhancement, hyperintensity as seen on Fluid Attenuated Inversion Recovery (FLAIR) was also measured. Pearson correlation statistics were performed to determine an association between insulin dosing and tumor or FLAIR growth velocity.

For each correlation analyzed (age, tumor size, FLAIR hyperintensity, average tumor growth velocity, and average FLAIR growth velocity versus survival and average or cumulative insulin dose), Pearson correlation coefficients and correlation significance were obtained (R function: *cor(pearson)*). Lines of best fit (including r^2 and slope values) were also obtained (R function: *abline*).

Other notable R graphics packages used included: *gtsummary*, *ggtext*, *ggplot2*, *scales*, *patchwork*, *tidyverse*, *forecasts*, *broom*, *tidyr* and *purrr*.

Results

There are some demographic differences between our diabetic (n=142) and non-diabetic (n=715) glioblastoma populations (Table 1 and Figure 1). Compared to non-diabetics, the median age of diabetics was older, 70 (range 39-93) years compared to 64 (range <1-93) years (Wilcoxon rank sum $p < 0.0001$). This is due to a greater proportion of diabetics identified at age 61 to 80 and >80 years (Chi-squared $p = 0.0013$). Furthermore, they had a lower median KPS compared to non-diabetics ($p = 0.0003$) and a higher percentage of them did not undergo resection, 39% compared to 32%, respectively, but this difference did not reach statistical significance ($p = 0.2291$). There is no difference in the number of males and females ($p = 0.4187$), treatment(s) received ($p = 0.2971$), *MGMT* promoter methylation status ($p = 0.2474$), and categorical Ki-67 index ($p = 0.1740$). Together, these findings support the notion that diabetes is a disease of aging, and glioblastoma patients with diabetes had poorer performance status and probably fewer of them underwent aggressive resections because of this.

Table 1: Demographic data and univariate analysis on non-diabetic and diabetic patients.

Variable	Non-Diabetics n = 715 ¹	Diabetics n = 142 ¹	p value
Age (years)			0.0013 ²
<40	35 (4.9%)	1 (0.7%)	
41-60	236 (33%)	31 (22%)	
61-80	380 (53%)	89 (63%)	
>80	64 (9.0%)	21 (15%)	
Median	64 (<1, 93)	70 (39, 93)	<0.0001 ³
Sex			0.4187 ²
Male	412 (58%)	76 (54%)	
Female	303 (42%)	66 (46%)	
Baseline KPS			0.0003 ²
Unknown	0 (0%)	2 (1.4%)	
0-60	235 (33%)	60 (42%)	
>80	244 (34%)	50 (35%)	
90-100	236 (33%)	30 (21%)	
Median	80 (0, 100)	70 (0, 100)	0.0015 ³
Resection Status			0.2291 ²
None	231 (32%)	56 (39%)	
STR	234 (33%)	39 (27%)	
GTR	250 (35%)	47 (33%)	
MGMT Status			0.2474 ²
Unmethylated	181 (25%)	31 (22%)	
Unknown	308 (43%)	72 (51%)	
Methylated	226 (32%)	39 (27%)	
Ki-67 Index			0.1740 ²
Unknown	1 (0.1%)	0 (0%)	
0-20%	288 (40%)	69 (49%)	
21-40%	217 (30%)	45 (32%)	
41-60%	148 (21%)	23 (16%)	
61-80%	56 (7.8%)	4 (2.8%)	
81-100%	5 (0.7%)	1 (0.7%)	
Median	30 (0, 100)	25 (0, 90)	0.0180 ³
Treatment Status			0.2971 ²
None	119 (17%)	31 (22%)	
RT Only	24 (3.4%)	7 (4.9%)	
Chemotherapy Only	7 (1.0%)	1 (0.7%)	
TMZ/RT Only	157 (22%)	35 (25%)	
TMZ/RT & 1+ Line	408 (57%)	68 (48%)	

¹ n (%); Median (Min, Max)² Pearson's Chi-squared test³ Wilcoxon rank sum test

KPS = Karnofsky Performance Scale; STR = Subtotal Resection; GTR = Gross Total Resection

MGMT = O⁶-methylguanine-DNA methyltransferase; RT = Radiation Therapy; TMZ = Temozolomide

Figure 1

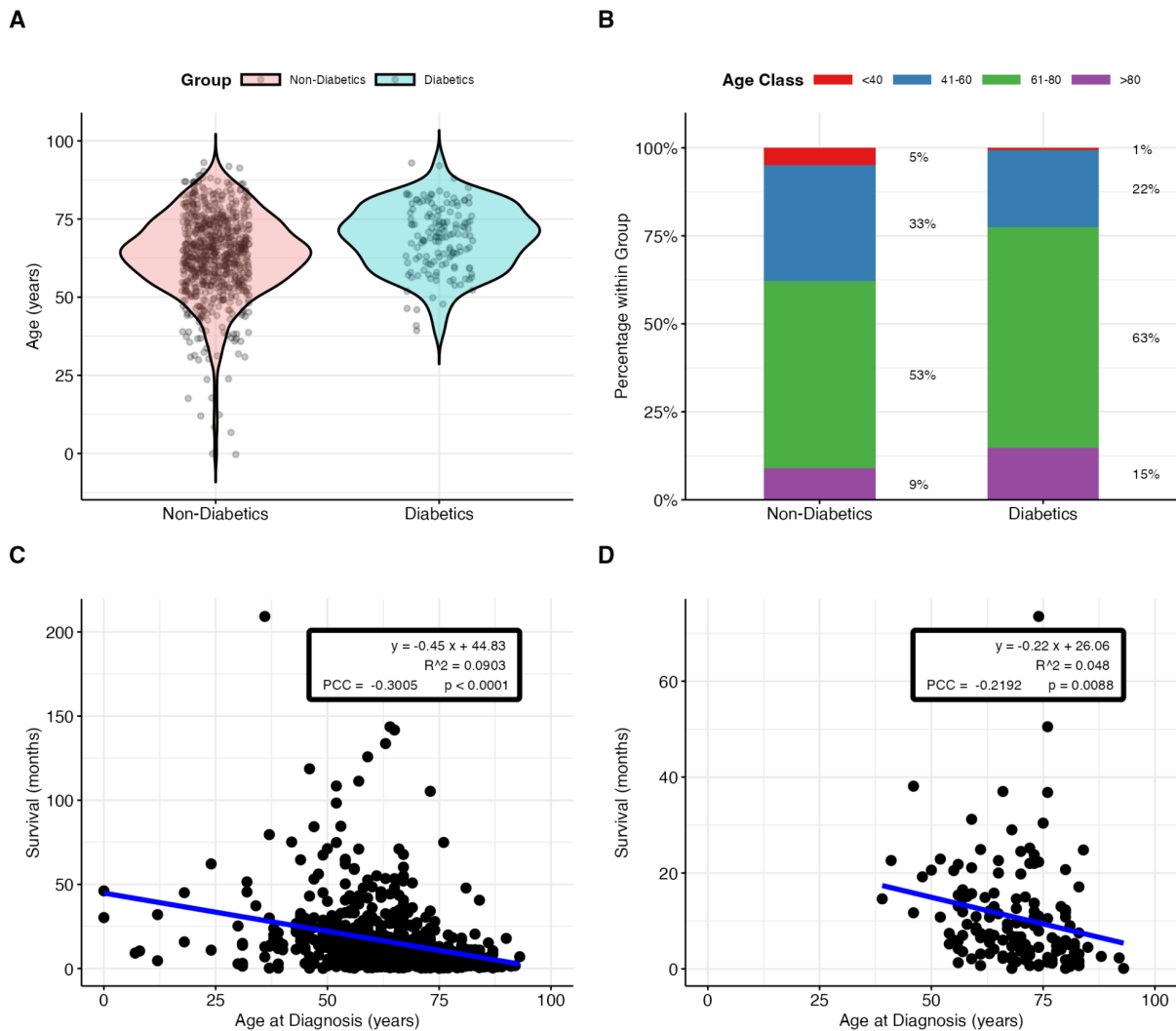


Figure 1: Age and survival in non-diabetic and diabetic patients with glioblastoma. Age distribution and median age between non-diabetics and diabetics (A). Categorical age distribution in non-diabetics according to age <40, 41-60, 61-80 and >80 years (B). Pearson correlation between age and survival in non-diabetics (C) and diabetics (D).

Glioblastoma patients with diabetes had shortened survival compared to the rest of the population

Compared to non-diabetics, glioblastoma patients with diabetes had a significantly shortened survival. Median OS was 7.9 (95%CI 6.4-10.8) months in diabetics compared to 11.9 (95%CI 11.0-13.0) months in non-diabetics ($p < 0.0001$) (Figure 2A). This prompted us to

perform an in-depth analysis of the influence of diabetes on glioblastoma patient survival. At the time of data extraction, all 142 patients were dead or censored.

Figure 2

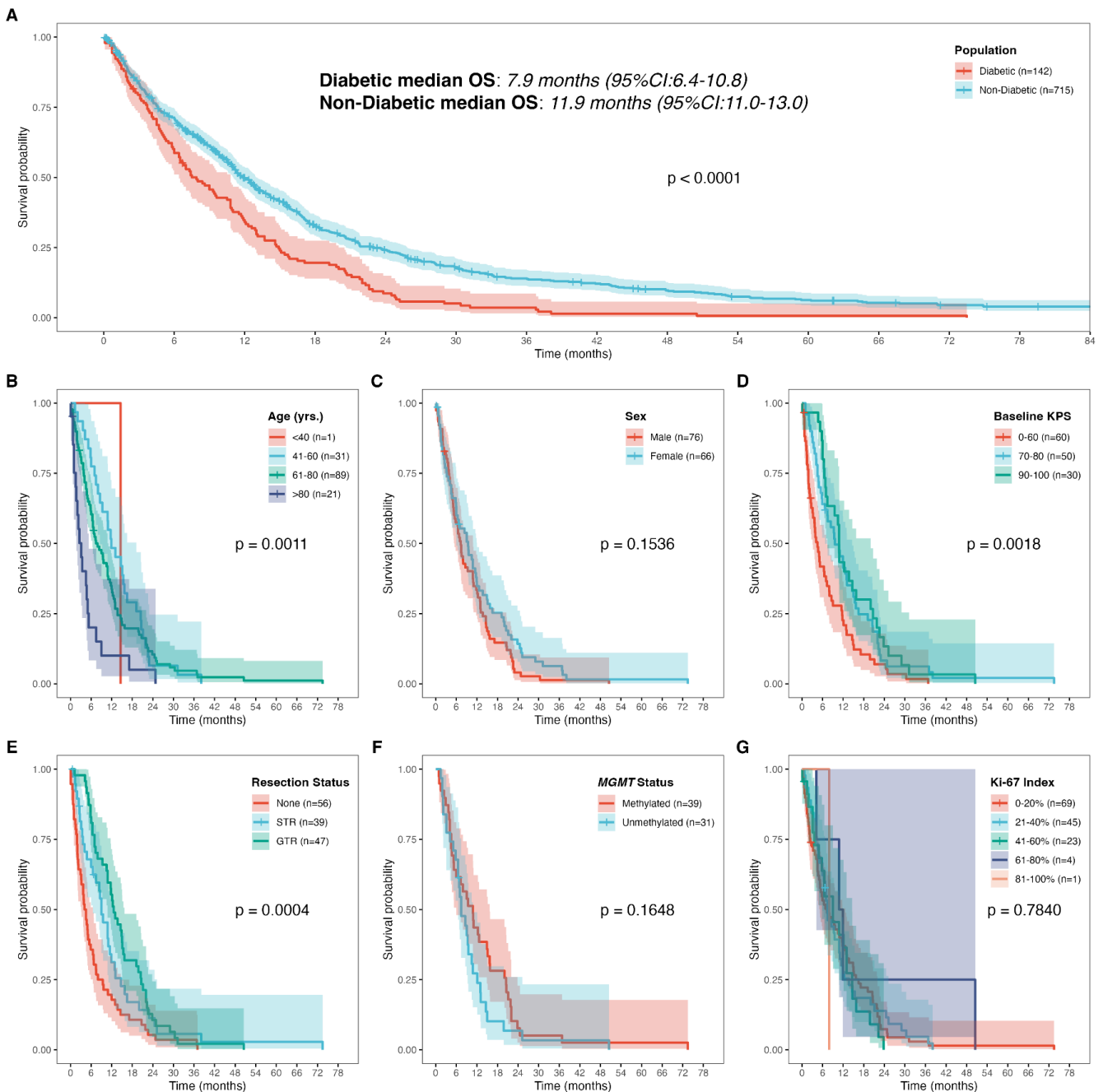


Figure 2: Kaplan-Meier survival and prognostic factors in non-diabetic and diabetic patients with glioblastoma. Non-diabetics had a longer median OS than diabetics (A). Among the diabetics, age (B), baseline KPS (D) and resection status (E) were significant prognostic factors, but not sex (C), MGMT promoter methylation status (F), or Ki-67 index (G).

Univariate and multivariate analyses revealed prognostic factors that correlate with overall survival

Our diabetic and non-diabetic cohorts have the expected prognostic factors for glioblastoma. For diabetics, univariate analyses showed expected survival correlation with age (when stratified into increments from <40, 40-60, 61-80, and >80 years), baseline KPS, and surgical resection status, but not sex or Ki-67 (Figure 2B-G) ⁵. Multivariate Cox proportional hazard analysis in the non-diabetic cohort showed significant correlations between survival and age, KPS, extent of surgical resection, and *MGMT* promoter methylation status (Table 2). Among the diabetics, KPS and resection status were significantly favoring those with better neurologic performance and gross total resections. There was a trend for increased hazard with older age and decreased hazard for those with methylated *MGMT* (Table 2) ^{18,19}. No specific treatment had an influence on survival (Table 2).

Table 2: Multivariate survival analysis of prognostic factors on non-diabetic and diabetic patients.

Prognostic Factor	Non-Diabetics			Diabetics		
	HR	95% CI	p value	HR	95% CI	p value
Age at Diagnosis (Continuous)	1.04	1.03, 1.04	<0.0001	1.02	1.00, 1.03	0.0661
Baseline KPS (Continuous)	0.99	0.99, 1.0	<0.0001	0.98	0.97, 0.99	<0.0001
Resection Status						
None	—	—		—	—	
STR	0.71	0.58, 0.87	0.0009	0.58	0.36, 0.92	0.0194
GTR	0.61	0.49, 0.75	<0.0001	0.46	0.30, 0.72	0.0006
MGMT Status						
Unmethylated	—	—		—	—	
Unknown	0.72	0.58, 0.89	0.0028	0.69	0.43, 1.11	0.1262
Methylated	0.56	0.45, 0.71	<0.0001	0.65	0.39, 1.09	0.1015
Treatment Status						
None	—	—		—	—	
RT Only	0.53	0.33, 0.84	0.0073	1.19	0.49, 2.89	0.7032
Chemotherapy Only	0.53	0.23, 1.22	0.1349	0.86	0.11, 6.64	0.8861
TMZ/RT Only	0.30	0.23, 0.39	<0.0001	0.78	0.47, 1.29	0.3320
TMZ/RT & 1+ Line	0.15	0.11, 0.20	<0.0001	1.00	0.64, 1.58	0.9928

Abbreviations: HR = Hazard Ratio; CI = Confidence Interval; KPS = Karnofsky Performance Status; MGMT = O⁶-methylguanine-DNA methyltransferase; STR = Subtotal Resection; GTR = Gross Total Resection; RT = Radiation Therapy; TMZ = Temozolomide.

Insulin use compromises glioblastoma survival

Insulin is a *bona fide* cellular growth factor and its cancer-promoting properties have been hypothesized previously^{20,21}. We therefore proceeded to compare the Kaplan-Meier survival characteristics of diabetic patients who used insulin and non-insulin medications, compared to those who did not have diabetes, and noted a significant difference among these groups (log rank $p < 0.0001$) (Figure 3A). Diabetic patients who used insulin ($n=122$) had a significantly compromised median OS compared to non-diabetics ($n=715$): 7.0 (95%CI 5.8-9.4) versus 11.9 (95%CI 11.1-13.2) months, respectively ($p < 0.0001$) (Figure 3C). Furthermore, insulin-using diabetics ($n=122$) also had a significantly reduced median OS compared to non-insulin-using diabetics ($n=20$): 7.0 (95% CI 5.8-9.4) versus 15.4 (95%CI 11.5-30.4) ($p=0.0109$) (Figure 3D). However, there was no difference between the non-diabetics ($n=715$) and non-insulin-using diabetics ($n=20$) ($p=0.9915$) (Figure 3E). Cox proportional hazard analysis showed a significantly increased hazard ratio in insulin-using but not non-insulin-using diabetics, with respective hazard ratios of 1.39 (95%CI 1.13-1.70) ($p=0.0015$) and 0.65 (95%CI 0.41-1.03) ($p=0.653$), when adjusted using covariate (age, KPS and resection status) found to be significant in our diabetic cohort (Table 2 and Figure 3B, upper panel). Alternatively, when using the significant covariates for the non-diabetic cohort (age, KPS, resection status, *MGMT* methylation and treatment)⁵ (Table 2), the hazard ratio became significant in the non-insulin-using but not in the insulin-using diabetics, hazard ratio 0.40 (95%CI 0.25-0.64) ($p=0.0001$) and 1.09 (95%CI 0.88-1.35) ($p=0.4195$), respectively (Figure 3B, lower panel). These findings suggest a trend for increased hazard disfavoring insulin use among glioblastoma patients with diabetes, but the sample size in either cohort is not large enough to differentiate the two groups definitively.

Figure 3

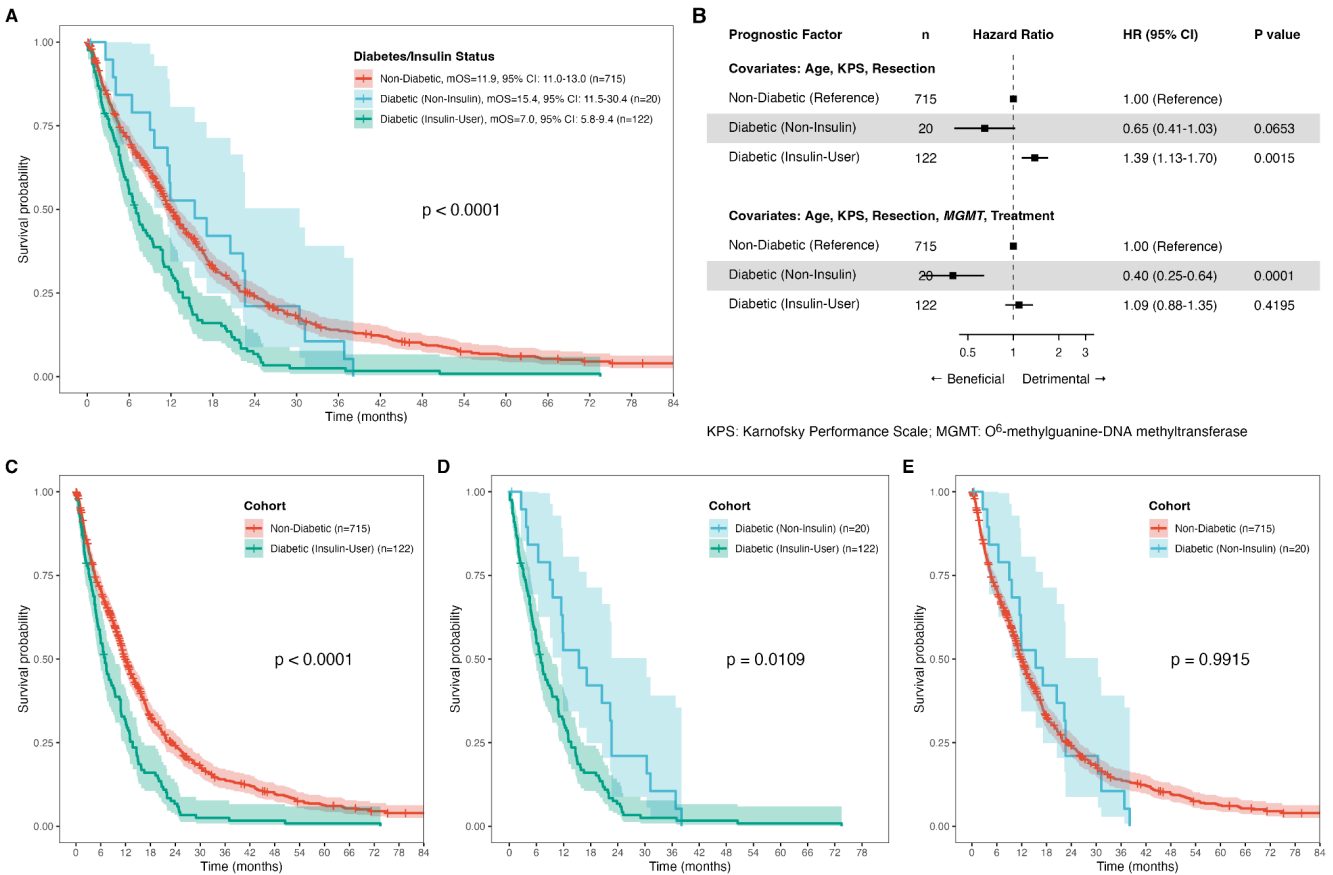


Figure 3: Insulin usage in diabetic patients with glioblastoma. Kaplan-Meier survival among non-diabetics, non-insulin-using diabetics, and insulin-using diabetics (**A**). Cox proportional hazard ratio of all diabetics, non-insulin-using diabetics and insulin-using diabetics, adjusted using significant covariates for diabetics (upper panel: age, KPS and resection status) and those significant for non-diabetics (lower panel: age, KPS, resection status, *MGMT* status and treatment) (**B**). Insulin-using diabetics had a shortened survival compared to non-diabetics (**C**) and non-insulin-using diabetics (**D**). There was no survival difference between non-diabetics and non-insulin-using diabetics (**E**).

We next query medication effects on our diabetics and divided them into 4 subgroups, with those used (i) insulin only (n=34), (ii) non-insulin diabetes medication(s) (n=8), (iii) both (n=88), or (iv) neither (n=12), and noted a statistical difference among them (log rank $p < 0.0001$) (Figure 4A). The non-insulin medications include glyburide, glimepiride, glipizide, sitagliptin, linagliptin, metformin, rosiglitazone, pioglitazone, acarbose, canagliflozin, empagliflozin, ertugliflozin, colesevelam, dulaglutide, exenatide, liraglutide, or any of the combinations. The median OS of patients who used insulin alone was 6.6 (95%CI 3.2-10.8) months which is

significantly shorter compared to those using non-insulin diabetes medication(s), 11.9 (95%CI 9.6-NR) months ($p=0.0357$), or no diabetes medication 17.1 (95%CI 11.5-NR) months ($p=0.0209$) (Figure 4C and 4D). There was a trend toward shortened survival among those who use both insulin and non-insulin diabetes medication(s), median OS 7.4 (95%CI 5.8-10.8), compared to those using neither, median OS 17.1 (95%CI 11.5-NR) ($p=0.0771$) (Figure 4E), or only non-insulin medication(s) 11.9 (95%CI 9.6-NR) ($p=0.1080$) (Supplementary Figure 1A). Cox proportional hazard analysis showed increased hazard ratios for those who used insulin with or without concurrent non-insulin diabetes medications, when adjusted using covariates significant for diabetics; in contrast, those who used only non-insulin diabetes medication(s) or neither had a decreased hazard ratio when adjusted using covariates significant for non-diabetics (Figure 4B). Compared to those taking non-insulin diabetes medications, insulin usage appears to correlate with a shortened survival in our diabetic cohort. Furthermore, no survival difference was noted between patients who used non-insulin diabetes medication(s) compared to those not using any medication at all ($p=0.7919$) (Supplementary Figure 1B), as well as patients who use insulin compared to those using both insulin and non-insulin medication(s) ($p=0.2573$) (Supplementary Figure 1C). Taken together, insulin use, either alone or in combination with other diabetes medication(s), appears to have a negative impact on glioblastoma patient survival.

Figure 4

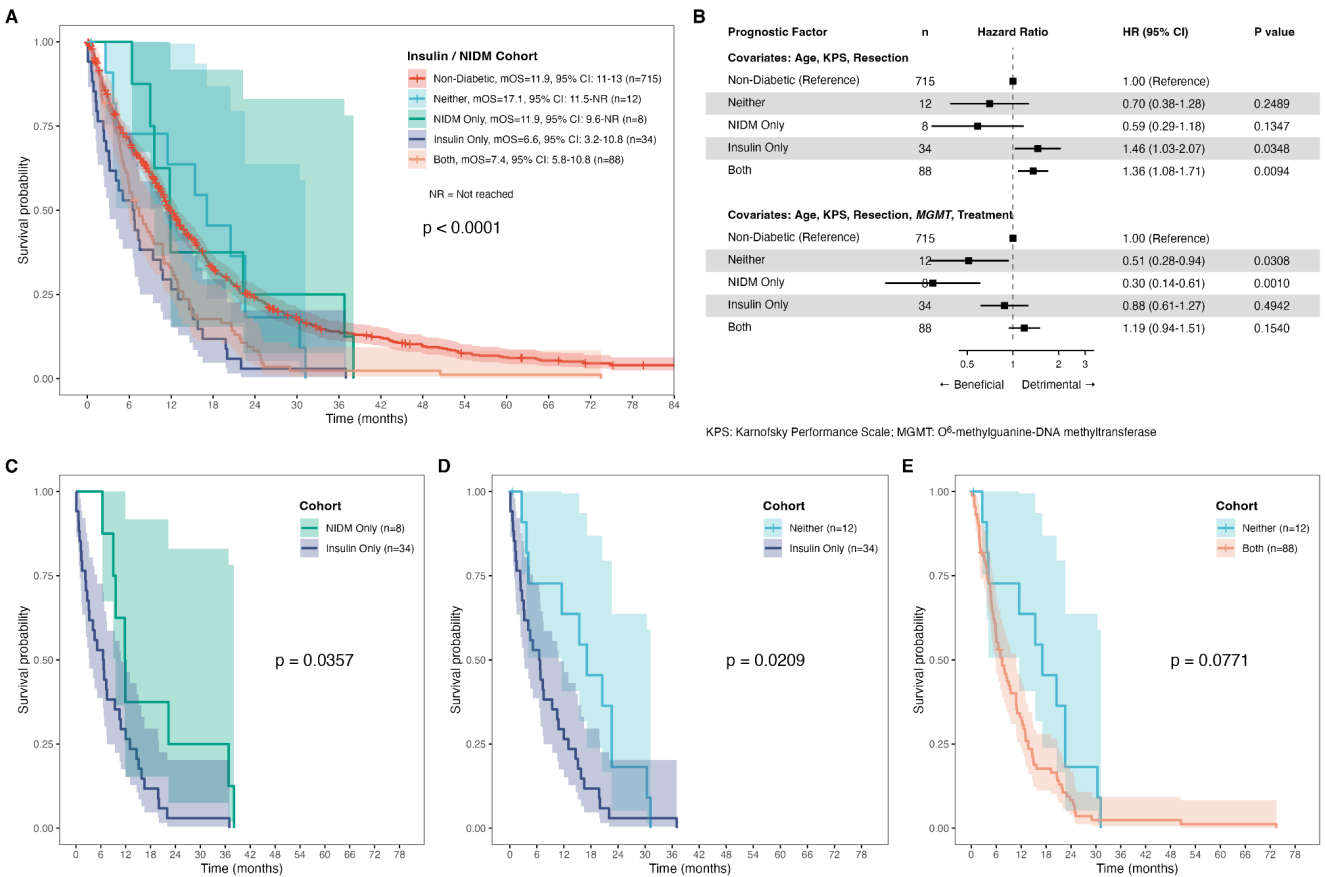


Figure 4: The influence of insulin versus non-insulin diabetes medication (NIDM) on patient survival. Kaplan-Meier survival showed a difference among patients using insulin, NIDM, both and neither (**A**). Cox proportional hazard ratio among patients using insulin, NIDM, both or neither, adjusted using significant covariates for diabetics (upper panel: age, KPS and resection status) and those significant for non-diabetics (lower panel: age, KPS, resection status, *MGMT* status and treatment) (**B**). Shortened survival was observed in patients using insulin compared to those using NIDM (**C**) or neither (**D**). There is a trend favoring patients using neither over those using both insulin and NIDM (**E**).

Shorter survival is associated with insulin usage after diagnosis and higher doses

We next investigated the correlation between patient survival and insulin usage before and/or after glioblastoma diagnosis. Survival analysis showed significant difference between non-diabetics and diabetics not using insulin (-/-, n=20) as well as those using insulin before and after diagnosis (+/+, n=51), only after diagnosis (-/+, n=68), and before but not after diagnosis (+/-, n=2) (log rank p<0.0001) (Figure 5). Compared to non-insulin-using (-/-) patients with a median OS of 15.4 (95%CI 11.5-30.4) months, those who used insulin (+/+ and -/+) had

shortened median OS of 5.8 (95%CI 4.7-7.9) months ($p=0.0092$) and 9.4 (95%CI 6.5-12.0) months ($p=0.0161$), respectively (Figures 5C and 5D). However, no survival difference was noted between post-diagnosis insulin use (-/+) and pre- and post-diagnosis insulin use (+/+) ($p=0.2782$) (Figure 5E). Pre-diagnosis insulin only patients (+/-), median OS 12.7 (95%CI 0.1-NR), were not analyzed due to a small sample size ($n=2$) preventing any meaningful comparison. Cox proportional hazard analysis showed significance in the +/+ group and a trend in the -/+ group for increased hazard when using covariates significant for the diabetic patients. In contrast, significance for decreased hazard was noted in the -/- group when using covariates significant for the non-diabetic patients. Taken together, insulin use before and/or after glioblastoma diagnosis appears to have a significant impact on patient survival.

Figure 5

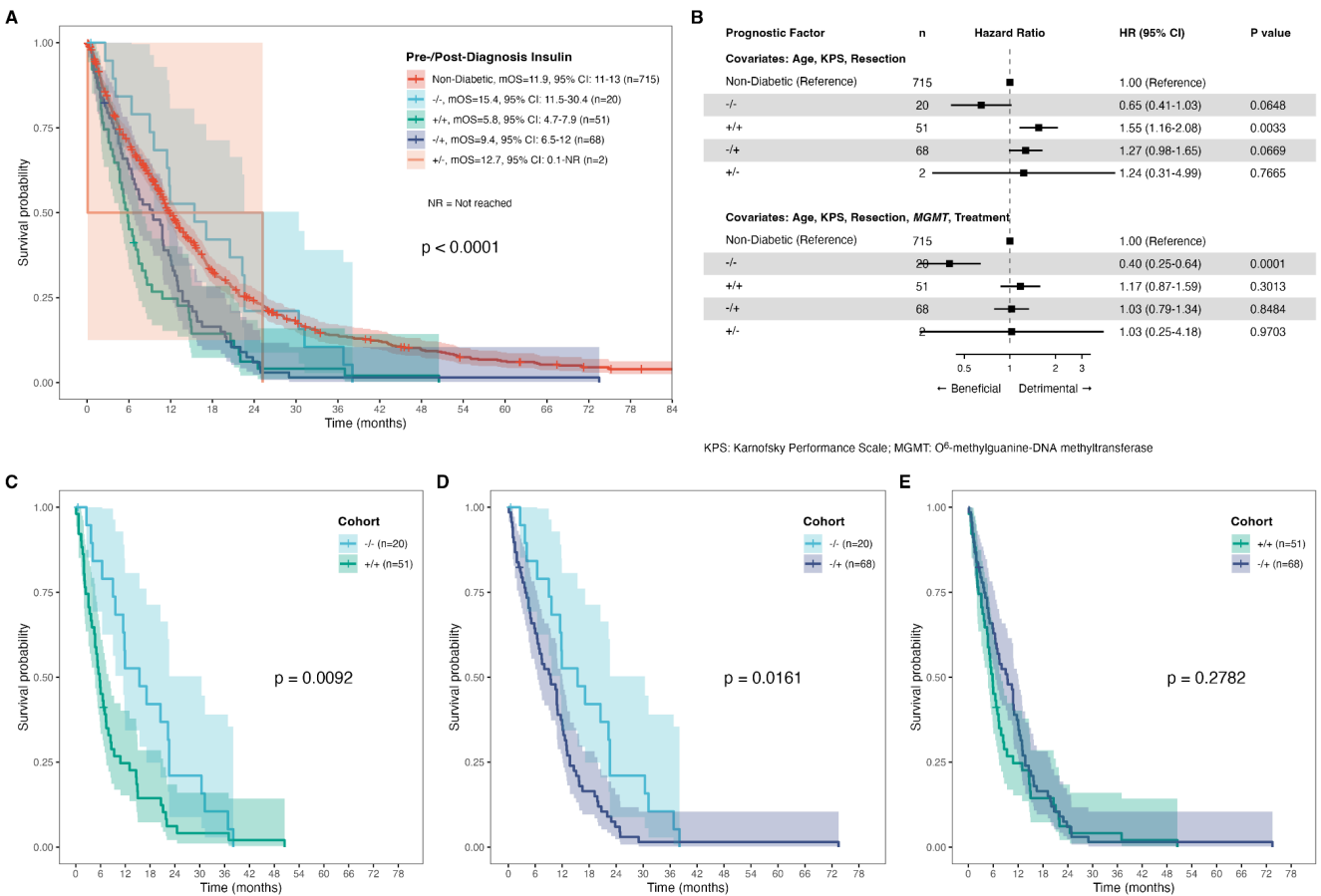


Figure 5: Pre- and post-diagnosis use of insulin among diabetics. Kaplan-Meier survival among diabetic patients who did not use insulin (-/-) or used before and after (+/+), only after (-/+) or only before diagnosis (+/-) (**A**). Cox proportional hazard ratio of -/-, +/+, -/+ or +/- groups, adjusted using significant covariates for diabetics (upper panel: age, KPS and resection status) and those significant for non-diabetics (lower panel: age, KPS, resection status, *MGMT* status and treatment) (**B**). Diabetics who did not use insulin (-/-) had significantly prolonged survival compared to those who used before and after (+/+) (**C**) and those who used only after (-/+) diagnosis (**D**). But there was no survival difference between those who used insulin after (-/+) and before and after (+/+) diagnosis (**E**).

Insulin usage could be a result of more severe diabetes in our glioblastoma population and therefore we analyzed the complication rates between insulin-using and non-insulin-using diabetics. These complications include retinopathy, nephropathy, neuropathy, foot ulcers, gangrene, infections requiring amputation, strokes, transient ischemic attacks, vascular dementia, coronary artery disease, heart attack, heart failure, peripheral vascular disease, and phlebitis. Wilcoxon rank sum analysis revealed no difference in the distribution on the number of complications among individual patients ($p=0.6009$) and categorical Chi-squared analysis

was also not significant ($p=0.8149$) (Supplementary Figure 2A). However, the level of glycosylated hemoglobin A1c was higher among patients using insulin ($p=0.0231$) (Supplementary Figure 2B). Taken together, our insulin-using patients probably have more advanced insulin resistance compared to those not using insulin, but the extent of end organ damage is similar between these two groups.

We then asked whether insulin use could be related to corticosteroid usage. The distribution of dexamethasone dosage was only marginally higher in the insulin-using patients compared to the non-insulin using ones (Wilcoxon rank sum test $p=0.0534$), and Chi-squared analysis of high versus low dose use (cutoff at 4 mg daily) was not significant ($p=0.1541$) (Supplementary Figure 2C).

We next explored the relationship between insulin dose and diabetic patient survival. Fifty-six (39%) patients did not have insulin dosage data in the medical records. In the remaining 86 patients, we determined the cutoffs for average daily insulin dosage and cumulative insulin use that lent itself to the greatest survival difference and the lowest p value between the two dichotomized cohorts. For average daily insulin dose, we found that this cutoff was near 10 units/day (Figure 6A). We then dichotomized the diabetic population based on the average daily insulin usage into (i) high-dose (>10 units/day) and (ii) low-dose (<10 units/day). Kaplan-Meier survival statistics showed a significant difference between these 2 subgroups. Patients who used high-dose insulin on a daily basis had a markedly shortened survival compared to those who used low dose: median OS 5.2 (95%CI 4.1-9.3) versus 10.5 (95%CI 7.4-13.1) months, respectively ($p=0.0006$). To guard against inadvertent type I error from multiple testing, we performed permutation analysis on our dichotomized insulin doses. As expected, the average daily dose of insulin near 10 unit/day was still significant ($p=0.0170$) (Figure 6B). Similarly, the cutoff for cumulative insulin dose was near 50 units/day (Figure 6C). Patients who used high cumulative doses of insulin also had a statistically significant shortened

survival compared to those who received low cumulative doses: median OS 6.7 (95%CI: 5.2-9.4) versus 11.5 (95%CI 7.4-21.8) months ($p=0.0104$). However, permutation analysis did not show significance at the cutoff dosage ($p=0.2290$). Collectively, these data indicate that insulin's negative effect on glioblastoma patient survival is dose-dependent and it is inversely related to the daily dose administered.

Figure 6

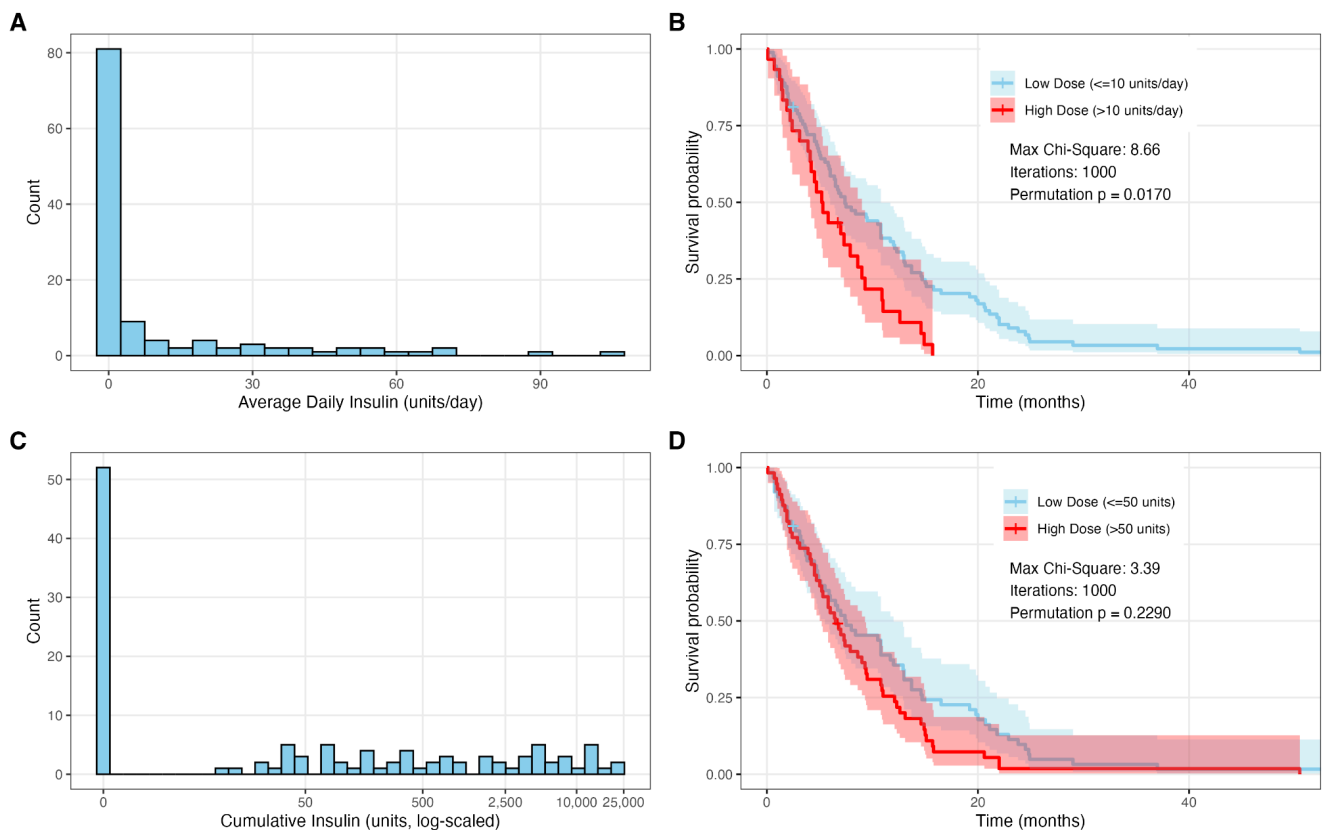


Figure 6: Low versus high insulin usage. Distribution of average daily insulin among diabetic patients (A). Kaplan-Meier survival difference between low versus high daily insulin usage at the optimal cutoff of 10 units/day, and survival difference remained significant after permutation testing with 1,000 iterations (B). Distribution of cumulative insulin among diabetic patients (C). Kaplan-Meier survival difference between low versus high cumulative insulin usage at the optimal cutoff of 50 units, but survival difference was not significant after permutation testing with 1,000 iterations (D).

Lack of association between insulin dosage and glioblastoma tumor size or growth velocity

We subsequently asked whether high insulin usage would correlate with tumor size or the growth velocity of glioblastoma. This analysis excluded an additional 51 patients because they (i) did not take insulin, (ii) did not have an initial resection, (iii) had ≤ 1 MRI scan archived at our institution, or (iv) did not have an available immediate post-resection MRI. Furthermore, to make comparisons uniform, we excluded tumor velocity measurements after a second resection. In the remaining subset of 35 patients, we found no correlation between the average daily insulin dose and tumor size as measured using the gadolinium-enhanced T1-weight MP RAGE (Pearson $r^2=0.0025$, $p=0.6846$) (Supplementary Figure 3A) or FLAIR images (Pearson $r^2<0.0001$, $p=0.9548$) (Supplementary Figure 3B). Similarly, there was no correlation between the cumulative insulin dose and tumor size as estimated using the gadolinium-enhanced T1-weight MP RAGE (Pearson $r^2=0.0011$, $p=0.7859$) (Supplementary Figure 3C) or FLAIR images (Pearson $r^2=0.0007$, $p=0.8254$) (Supplementary Figure 3D).

Glioblastoma growth velocity was computed by determining the difference in tumor size between first post-resection MRI and each of the subsequent MRI scans divided by the time interval between them. No significant Pearson correlation was noted between daily insulin dose and tumor size (Supplementary Figure 3E). Although there is a slight but significant downward sloping relationship between daily insulin dose and FLAIR images (Supplementary Figure 3F), this may be just a stochastic anomaly since the correlation was insignificant between the cumulative insulin dose and tumor size (Supplementary Figure 3G) and FLAIR images (Supplementary Figure 3H). Alternatively, the downward sloping could be from decreased inflammation as a result of normalized glucose in the interstitium.

Discussion

An important finding from our analysis is that insulin use has an overriding negative survival impact on our glioblastoma patients, and this relationship is probably dose dependent. The survival of diabetics in our cohort who used insulin was shortened by more than one-third compared to those who did not. Furthermore, there was no survival difference between non-diabetics and diabetics who did not use insulin. Prior retrospective studies did not focus on insulin utilization, probably due to the significant fluctuations in dose administered, and the emphasis was on the presence or absence of T2DM or the severity of hyperglycemia in these patients ^{13,22}. Importantly, fast-acting insulin is the quickest and most effective way to ameliorate hyperglycemia ²³ and, when patients are placed on insulin, few are switched to non-insulin secretagogues, GLP-1 agonist or other drugs. However, we meticulously extracted the dosage from electronic medical records and identified the average daily dose and the cumulative dose for each patient. Our effort revealed that there appears a dose-dependent relationship between survival and daily insulin dosage but not with cumulative insulin use. This finding is very similar to our prior *post hoc* clinical trial analysis of dexamethasone effect on recurrent glioblastoma patients randomized to Tumor Treating Fields or chemotherapy, in which daily doses are more correlative with survival than cumulative doses ²⁴. If this is confirmed in future studies, glycemic control without using insulin or with non-insulin diabetic medications may be a judicious approach for managing T2DM in glioblastoma patients.

Derr et al. performed a retrospective analysis of hyperglycemia among newly diagnosed glioblastoma patients enrolled in 5 phase II clinical trials. They found that those with a time-weighted mean serum glucose level of >137 mg/dL had a significantly decreased survival, with a hazard ratio of 1.57, compared to those with a level of <94 mg/dL after controlling for corticosteroid use and other confounders ¹⁵. They posit that intensive management of corticosteroid-induced hyperglycemia may improve the longevity of these patients. Other studies also revealed an association with survival, at a serum glucose cutoff of 90 mg/dL ²⁵, 112

mg/dL ²⁶, 113 mg/dL ²², 167 mg/dL ²⁷ or 180 mg/dL ^{28,29}, while others did not find an association^{30,31}. However, it is still unclear whether poor survival associated with high levels of hyperglycemia in these patients is due to intensive insulin use, complications from T2DM, extent of dexamethasone utilization, or any of the combinations above ³².

A remarkable finding from our dataset is that diabetics had a shortened OS compared to non-diabetics. However, the association between T2DM and cancer patients is controversial, where some studies found an association^{29,31,33} while others did not ^{8,28, 30, 34–38}. Most of these studies did not investigate complications associated with T2DM or their contribution to patient survival. Regardless, there are definitely more negative than positive studies, suggesting that T2DM *per se* in the glioblastoma population may not attenuate their survival.

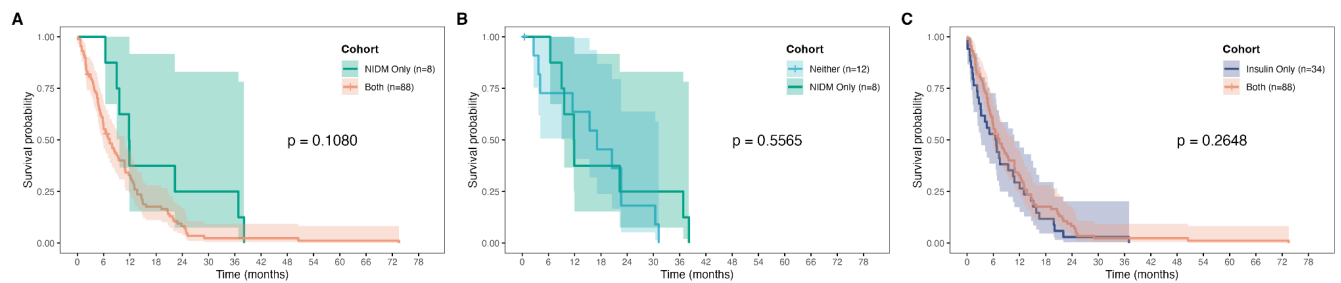
Insulin is a *bona fide* growth factor that binds to both insulin and insulin-like growth factor receptors, triggering cellular growth and proliferation signals via the phosphoinositide-3 kinase (PI3K) pathway ^{39,40}. This pathway also regulates metabolic functions in tissues and organs for homeostasis ⁴¹. According to The Cancer Genome Atlas, alterations in PI3K were found in 40% of glioblastoma, including amplifications as well as activating mutations and fusions ⁴¹. However, the brain-penetrant PI3K inhibitor, GDC-0084 (buparlisib), showed no efficacy in treating recurrent glioblastoma despite achieving pharmacodynamic concentrations in the patient's brain and the tumor ^{42,43}. This is most likely due to drug-induced hyperglycemia resulting in a secondary insulin feedback escape mechanism, and suppression of this insulin feedback in mouse models enhanced the efficacy of PI3K inhibitors ^{44,45}. The collective data indicate that PI3K inhibition alone is a useless therapeutic strategy for glioblastoma without blocking the possible secondary surge of endogenous insulin as a result of hyperglycemia or suppressing the use of exogenous insulin to normalize high serum glucose levels.

This study has several limitations. First, our patients are heterogeneous but they possess known prognostic factors that are typically found in the glioblastoma population. Second, it is difficult to resolve individual effects from dexamethasone use, administration of insulin, or hyperglycemia triggering compensatory endogenous hyperinsulinemia or administration of external insulin for lowering high serum glucose. However, our data suggest an association between insulin dose and survival, and the contribution to the latter from diabetic complications or dexamethasone use is minimal. Lastly, there is no correlation between insulin and tumor size. This lack of correlation is probably secondary to the infiltrative nature of glioblastoma, making precise measurement of tumor size difficult. The bi-dimensional tumor measurement, consisting of the product of two longest perpendicular diameters, has limitations in the RANO criteria ⁴⁶. Therefore, volumetric measurement of tumor size in RANO 2.0 probably provides a more accurate assessment ¹⁷, but it is still difficult to account for the microscopic glioblastoma cells that infiltrate into the adjacent brain.

In conclusion, our analysis reveals that our diabetic glioblastoma patients who used insulin had a significantly compromised survival compared to non-diabetics or non-insulin-using diabetics. Those who received higher daily insulin doses had a shorter survival than others who used lower doses. Therefore, in addition to administering cancer drug treatments, oncologists' attention to diabetes and insulin use will be paramount in the overall management of the glioblastoma population.

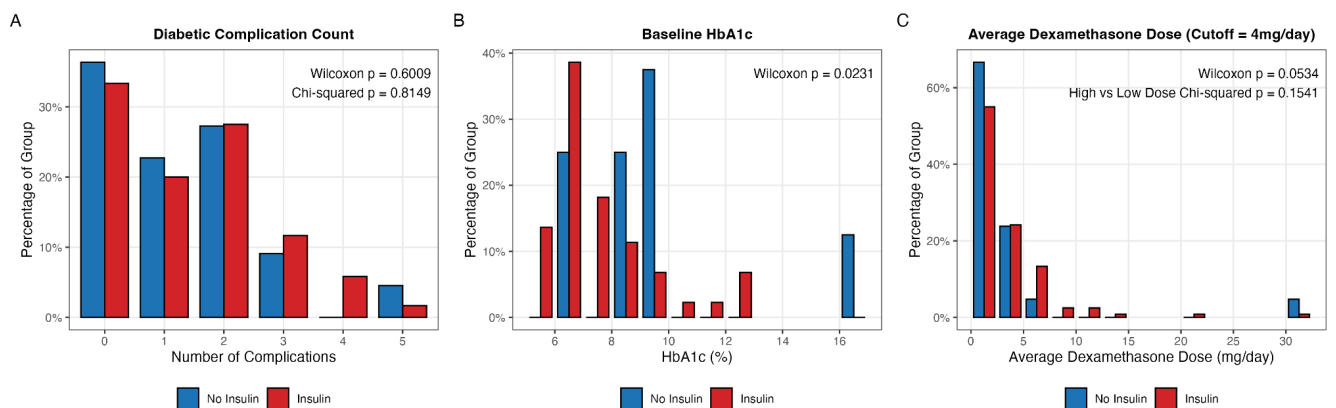
Supplementary Figures

Supplementary Figure 1



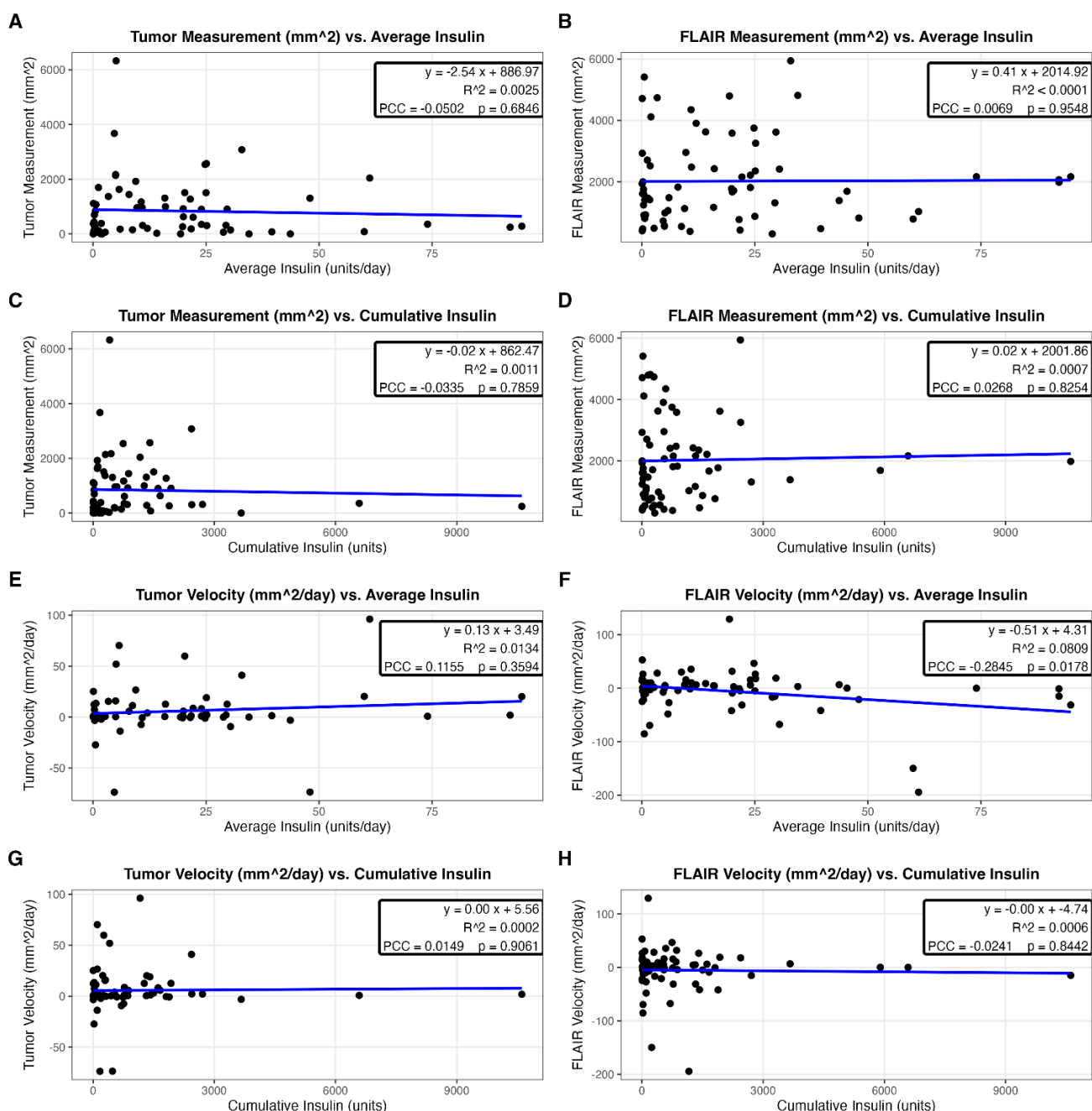
Supplementary Figure 1: Additional analysis on the influence of insulin versus non-insulin diabetes medications (NIDM) on patient survival. Compared to patients using both insulin and NIDM, there is a trend favoring those using NIDM only (A). No difference was seen in survival between patients using NIDM and neither (B), or between those using insulin only and both insulin and NIDM (D).

Supplementary Figure 2



Supplementary Figure 2: Severity of diabetic complications, serum marker and dexamethasone usage. There was no difference in the distribution of the number of diabetic complications between the non-diabetic and diabetic cohorts (A). A higher hemoglobin A1c was seen in diabetic patients than non-diabetics (B). Distribution of dexamethasone dosage in insulin-using patients was only marginally higher compared to non-insulin-using patients, and no difference was noted between high versus low dose dexamethasone use in the two groups (C).

Supplementary Figure 3



Supplementary Figure 3: Glioblastoma tumor size and growth velocity versus daily and cumulative insulin dose. No difference was seen between average daily insulin dose and bi-dimensional tumor measurements from T1-weighted post-gadolinium MP RAGE (**A**) and FLAIR (**B**) images. No difference was also observed between cumulative insulin dose and bi-dimensional tumor measurements from T1-weighted post-gadolinium MP RAGE (**C**) and FLAIR (**D**) images. No difference was noted between average daily insulin dose and tumor growth velocity as measured using T1-weighted post-gadolinium MP RAGE images (**E**). There is a slight but downward sloping relationship between average daily insulin dose and tumor growth velocity as measured using FLAIR images (**F**). No difference was seen between the cumulative insulin dose and tumor growth velocity as estimated using T1-weighted post-gadolinium MP RAGE (**G**) or FLAIR images (**H**).

References

1. Wirsching HG, Galanis E, Weller M. Glioblastoma. *Handb Clin Neurol*. 2016;134:381-397. doi:10.1016/B978-0-12-802997-8.00023-2
2. Nawabi NLA, Saway BF, Jha R, et al. Current Trends in the Allocation of NIH Funding of Brain Tumor Research. *Neuro-Oncol Adv*. Published online December 4, 2024:vdae203. doi:10.1093/noajnl/vdae203
3. Girardi F, Matz M, Stiller C, et al. Global survival trends for brain tumors, by histology: analysis of individual records for 556,237 adults diagnosed in 59 countries during 2000-2014 (CONCORD-3). *Neuro-Oncol*. 2023;25(3):580-592. doi:10.1093/neuonc/noac217
4. Chen W, Zhang T, Zhang H. Causal relationship between type 2 diabetes and glioblastoma: bidirectional Mendelian randomization analysis. *Sci Rep*. 2024;14(1):16544. doi:10.1038/s41598-024-67341-x
5. Bessette J, Arditi J, Noreck J, et al. Medical comorbidities prognosticate survival in glioblastoma. *BJC Rep*. 2025;3(1):75. doi:10.1038/s44276-025-00189-4
6. Wu Y, Ding Y, Tanaka Y, Zhang W. Risk factors contributing to type 2 diabetes and recent advances in the treatment and prevention. *Int J Med Sci*. 2014;11(11):1185-1200. doi:10.7150/ijms.10001
7. Suh S, Kim KW. Diabetes and Cancer: Cancer Should Be Screened in Routine Diabetes Assessment. *Diabetes Metab J*. 2019;43(6):733. doi:10.4093/dmj.2019.0177
8. Barami K, Lyon L, Conell C. Type 2 Diabetes Mellitus and Glioblastoma Multiforme-Assessing Risk and Survival: Results of a Large Retrospective Study and Systematic Review of the Literature. *World Neurosurg*. 2017;106:300-307. doi:10.1016/j.wneu.2017.06.164
9. Yoshikawa MH, Rabelo NN, Telles JPM, Figueiredo EG. Modifiable risk factors for glioblastoma: a systematic review and meta-analysis. *Neurosurg Rev*. 2023;46(1):143. doi:10.1007/s10143-023-02051-y
10. Zhao L, Zheng Z, Huang P. Diabetes mellitus and the risk of glioma: a meta-analysis. *Oncotarget*. 2016;7(4):4483-4489. doi:10.18632/oncotarget.6605
11. *National Diabetes Statistics Report*. 2024. <https://www.cdc.gov/diabetes/php/data-research/index.html>
12. Rooney MR, Fang M, Ogurtsova K, et al. Global Prevalence of Prediabetes. *Diabetes Care*. 2023;46(7):1388-1394. doi:10.2337/dc22-2376

13. Montemurro N, Perrini P, Rapone B. Clinical Risk and Overall Survival in Patients with Diabetes Mellitus, Hyperglycemia and Glioblastoma Multiforme. A Review of the Current Literature. *Int J Environ Res Public Health*. 2020;17(22):8501. doi:10.3390/ijerph17228501
14. Lucas D, Carvalho B, Tuna R, Linhares P. Metabolic Syndrome and Survival in Glioblastoma Patients: Retrospective Cohort Study and Review of the Literature. *Cureus*. 2024;16(2):e53641. doi:10.7759/cureus.53641
15. Derr RL, Ye X, Islas MU, Desideri S, Saudek CD, Grossman SA. Association between hyperglycemia and survival in patients with newly diagnosed glioblastoma. *J Clin Oncol Off J Am Soc Clin Oncol*. 2009;27(7):1082-1086. doi:10.1200/JCO.2008.19.1098
16. Davidson-Pilon C. lifelines: survival analysis in Python. *J Open Source Softw*. 2019;4(40):1317. doi:10.21105/joss.01317
17. Wen PY, Van Den Bent M, Youssef G, et al. RANO 2.0: Update to the Response Assessment in Neuro-Oncology Criteria for High- and Low-Grade Gliomas in Adults. *J Clin Oncol*. 2023;41(33):5187-5199. doi:10.1200/JCO.23.01059
18. Hegi ME, Diserens AC, Gorlia T, et al. *MGMT* Gene Silencing and Benefit from Temozolomide in Glioblastoma. *N Engl J Med*. 2005;352(10):997-1003. doi:10.1056/NEJMoa043331
19. Annavarapu S, Gogate A, Pham T, Davies K, Singh P, Robert N. Treatment patterns and outcomes for patients with newly diagnosed glioblastoma multiforme: a retrospective cohort study. *CNS Oncol*. 2021;10(3):CNS76. doi:10.2217/cns-2021-0007
20. Mårmol JM, Carlsson M, Raun SH, et al. Insulin resistance in patients with cancer: a systematic review and meta-analysis. *Acta Oncol Stockh Swed*. 2023;62(4):364-371. doi:10.1080/0284186X.2023.2197124
21. Vigneri R, Sciacca L, Vigneri P. Rethinking the Relationship between Insulin and Cancer. *Trends Endocrinol Metab TEM*. 2020;31(8):551-560. doi:10.1016/j.tem.2020.05.004
22. Tieu MT, Lovblom LE, McNamara MG, et al. Impact of glycemia on survival of glioblastoma patients treated with radiation and temozolomide. *J Neurooncol*. 2015;124(1):119-126. doi:10.1007/s11060-015-1815-0
23. Weng J. Short-term intensive insulin therapy could be the preferred option for new onset Type 2 diabetes mellitus patients with HbA1c > 9. *J Diabetes*. 2017;9(10):890-893. doi:10.1111/1753-0407.12581

24. Wong ET, Lok E, Gautam S, Swanson KD. Dexamethasone exerts profound immunologic interference on treatment efficacy for recurrent glioblastoma. *Br J Cancer*. 2015;113(2):232-241. doi:10.1038/bjc.2015.238
25. Stevens G, Ahluwalia M. Elevated Preoperative Glucose Levels and Survival in Elderly Newly Diagnosed Glioblastoma Patients. *Neurology*. Published online April 23, 2012. doi:P07-111
26. Hagan K, Bhavsar S, Arunkumar R, et al. Association Between Perioperative Hyperglycemia and Survival in Patients With Glioblastoma. *J Neurosurg Anesthesiol*. 2017;29(1):21-29. doi:10.1097/ANA.0000000000000339
27. Decker M, Sacks P, Abbatematteo J, De Leo E, Brennan M, Rahman M. The effects of hyperglycemia on outcomes in surgical high-grade glioma patients. *Clin Neurol Neurosurg*. 2019;179:9-13. doi:10.1016/j.clineuro.2019.02.011
28. McGirt MJ, Chaichana KL, Gathinji M, et al. Persistent outpatient hyperglycemia is independently associated with decreased survival after primary resection of malignant brain astrocytomas. *Neurosurgery*. 2008;63(2):286-291; discussion 291. doi:10.1227/01.NEU.0000315282.61035.48
29. Chambless LB, Parker SL, Hassam-Malani L, McGirt MJ, Thompson RC. Type 2 diabetes mellitus and obesity are independent risk factors for poor outcome in patients with high-grade glioma. *J Neurooncol*. 2012;106(2):383-389. doi:10.1007/s11060-011-0676-4
30. Seliger C, Genbrugge E, Gorlia T, et al. Use of metformin and outcome of patients with newly diagnosed glioblastoma: Pooled analysis. *Int J Cancer*. 2020;146(3):803-809. doi:10.1002/ijc.32337
31. Welch MR, Grommes C. Retrospective analysis of the effects of steroid therapy and antidiabetic medication on survival in diabetic glioblastoma patients. *CNS Oncol*. 2013;2(3):237-246. doi:10.2217/cns.13.12
32. Wong ET, Swanson KD. Dexamethasone-Friend or Foe for Patients With Glioblastoma? *JAMA Neurol*. 2019;76(3):247-248. doi:10.1001/jamaneurol.2018.4530
33. Chen YR, Ugiliweneza B, Burton E, Woo SY, Boakye M, Skirboll S. The effect of postoperative infection on survival in patients with glioblastoma. *J Neurosurg*. 2017;127(4):807-811. doi:10.3171/2016.8.JNS16836
34. Grommes C, Conway DS, Alshekhlee A, Barnholtz-Sloan JS. Inverse association of PPAR γ agonists use and high grade glioma development. *J Neurooncol*. 2010;100(2):233-239. doi:10.1007/s11060-010-0185-x

35. Siegel EM, Nabors LB, Thompson RC, et al. Prediagnostic body weight and survival in high grade glioma. *J Neurooncol*. 2013;114(1):79-84. doi:10.1007/s11060-013-1150-2
36. Mayer A, Vaupel P, Struss HG, Giese A, Stockinger M, Schmidberger H. Strong adverse prognostic impact of hyperglycemic episodes during adjuvant chemoradiotherapy of glioblastoma multiforme. *Strahlenther Onkol Organ Dtsch Rontgengesellschaft Al*. 2014;190(10):933-938. doi:10.1007/s00066-014-0696-z
37. Adeberg S, Bernhardt D, Ben Harrabi S, et al. Metformin influences progression in diabetic glioblastoma patients. *Strahlenther Onkol Organ Dtsch Rontgengesellschaft Al*. 2015;191(12):928-935. doi:10.1007/s00066-015-0884-5
38. Potharaju M, Mangaleswaran B, Mathavan A, et al. Body Mass Index as a Prognostic Marker in Glioblastoma Multiforme: A Clinical Outcome. *Int J Radiat Oncol Biol Phys*. 2018;102(1):204-209. doi:10.1016/j.ijrobp.2018.05.024
39. Haeusler RA, McGraw TE, Accili D. Biochemical and cellular properties of insulin receptor signalling. *Nat Rev Mol Cell Biol*. 2018;19(1):31-44. doi:10.1038/nrm.2017.89
40. Hakuno F, Takahashi SI. IGF1 receptor signaling pathways. *J Mol Endocrinol*. 2018;61(1):T69-T86. doi:10.1530/JME-17-0311
41. Hopkins BD, Goncalves MD, Cantley LC. Insulin-PI3K signalling: an evolutionarily insulated metabolic driver of cancer. *Nat Rev Endocrinol*. 2020;16(5):276-283. doi:10.1038/s41574-020-0329-9
42. Wen PY, Cloughesy TF, Olivero AG, et al. First-in-Human Phase I Study to Evaluate the Brain-Penetrant PI3K/mTOR Inhibitor GDC-0084 in Patients with Progressive or Recurrent High-Grade Glioma. *Clin Cancer Res Off J Am Assoc Cancer Res*. 2020;26(8):1820-1828. doi:10.1158/1078-0432.CCR-19-2808
43. Wen PY, Touat M, Alexander BM, et al. Buparlisib in Patients With Recurrent Glioblastoma Harboring Phosphatidylinositol 3-Kinase Pathway Activation: An Open-Label, Multicenter, Multi-Arm, Phase II Trial. *J Clin Oncol Off J Am Soc Clin Oncol*. 2019;37(9):741-750. doi:10.1200/JCO.18.01207
44. Hopkins BD, Pauli C, Du X, et al. Suppression of insulin feedback enhances the efficacy of PI3K inhibitors. *Nature*. 2018;560(7719):499-503. doi:10.1038/s41586-018-0343-4
45. Hopkins BD, Pauli C, Du X, et al. Publisher Correction: Suppression of insulin feedback enhances the efficacy of PI3K inhibitors. *Nature*. 2018;563(7731):E24. doi:10.1038/s41586-018-0506-3
46. Wen PY, Macdonald DR, Reardon DA, et al. Updated response assessment criteria for high-grade gliomas: response assessment in neuro-oncology working group. *J*

Clin Oncol Off J Am Soc Clin Oncol. 2010;28(11):1963-1972.
doi:10.1200/JCO.2009.26.3541